

Introduction to Vector Databases.

AI applications don't work on exact words or values, they work on the context/meaning.

Embeddings.

→ Numerical representation of data such as text, image, audio or video etc.

→ List of Numbers.

Vector.

Apple → [0.12 0.84 0.51 0.94 -0.11 ]

→ No. of elements in a Vector depends on the embedding model that is being used.

→ Embeddings are powerful as they captures semantic meaning.

⇒ Cheap flights from Delhi to Bangalore

⇒ Show me flights from Delhi to Bangalore which are affordable.

Similar meaning $\Rightarrow$ Close Vectors.

Different meaning $\Rightarrow$ Far Vectors.

Embeddings.

- Semantic Search
- Recommendation System.
- Clustering / Grouping / Classification.
- Anomaly / Fraud detection.

Why Traditional DBs are limited for Vector data?

↓
Relational DBs.
↳ MySQL / PostgreSQL / MSQL.

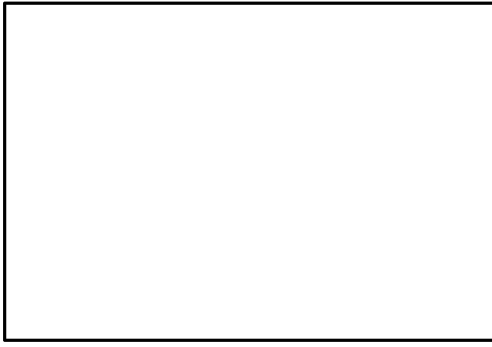
↳ Structured data.

↳ order-id, product-id, status, date, - - -

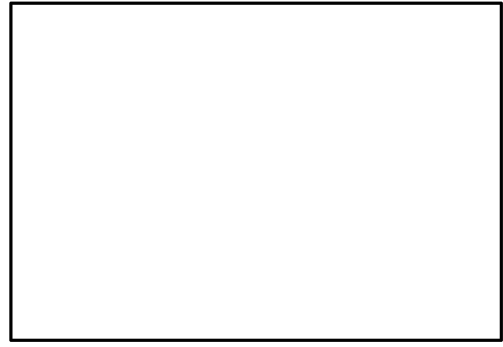
Queries.

- fetch order with id = 1234.
- fetch 10 latest orders for user with id = 10.
- JOIN queries.
- filter
- sorting.

Orders



Products.



Users



Queries in traditional DBs $\Rightarrow$ exact match

$\Rightarrow$ Traditional DBs

$\hookrightarrow$ Millions of Vectors.

1000s of dimensions.



Query: cheap flights from Delhi to Bangalore



5 nearest vectors

find the top 5 vectors similar to the query vector.

→ find the distance of each vector from the query vector, arrange them in sorted order & get top 5.

Vector Database.

→ It is designed to store, manage & search embeddings efficiently.

→ Stores data in the form of vectors & retrieve results based on similarity & not just on exact match.

→ Store embeddings.

→ Search embeddings based on similarity.

→ Scale to large collections.

→ Semantic retrieval.

Step-by-Step Approach.

→ Document Uploading

→ Generate Embeddings.

→ Store Embeddings in Vector DB.

→ Query.

→ Generate embedding for query.

→ Search nearest vector wrt query Vector.

⇒ While storing & retrieving embeddings, same model should be used.

Similarity Based Retrieval.

→ NO exact word search.

→ Search based on meaning / semantics.

Similarity Measurement.

Similar meaning $\rightarrow$ Close Vectors.

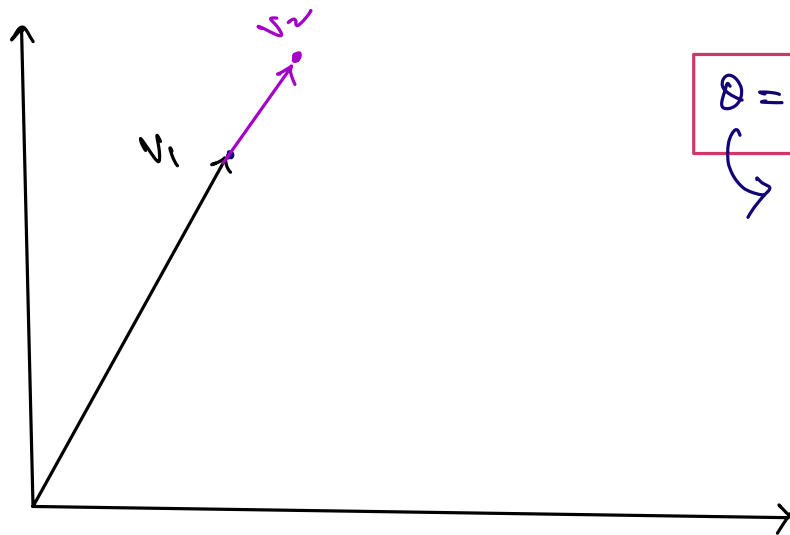
Different meaning $\rightarrow$ Far apart
Vectors.

How to measure the closeness of 2 vectors?

I) Cosine Similarity

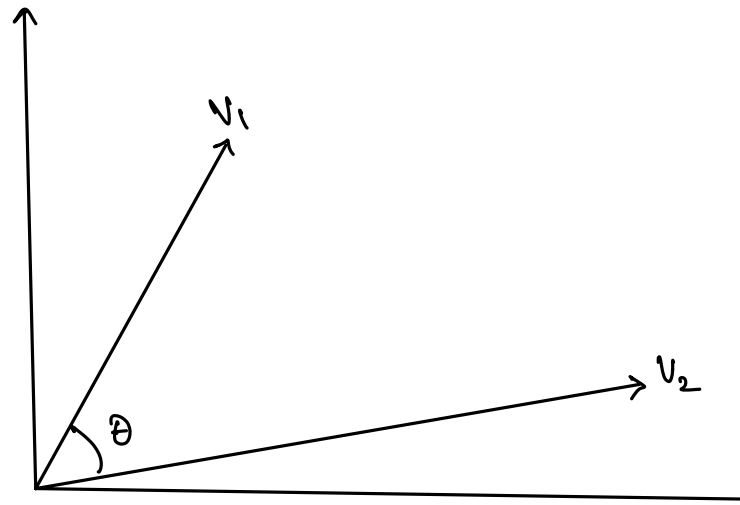
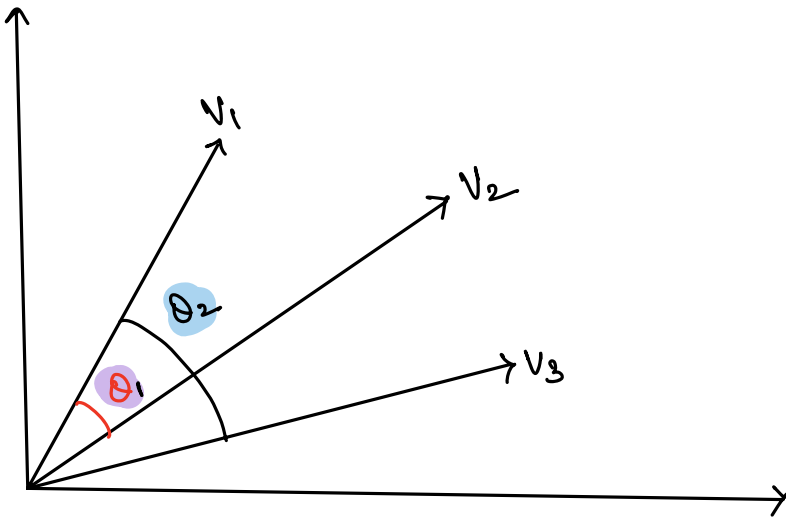
II) Euclidean Distance.

Cosine Similarity



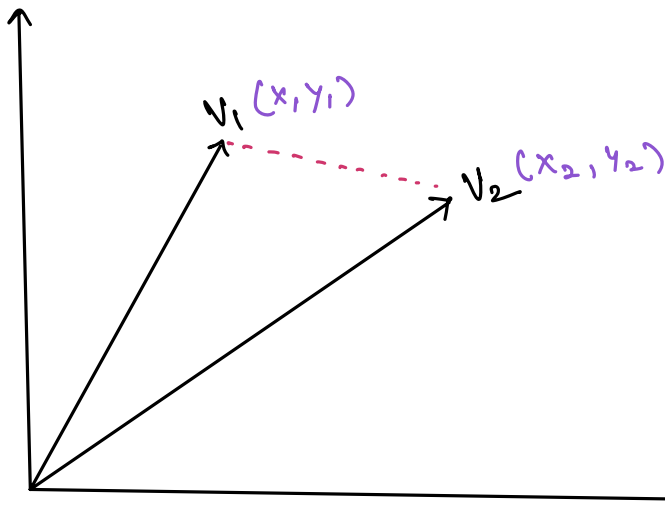
$$\theta = 0 \Rightarrow \cos \theta = 1$$

$\hookrightarrow$ More Similar.



Angle b/w the Vector $\uparrow$ $\Rightarrow$ Similarity $\downarrow$

Euclidean Distance.



distance $\uparrow$ $\Rightarrow$ Similarity $\downarrow$

Why is Scale Search needed?

Millions of documents.

Vector Indexing.

⇒ Data Structure that helps the system search similar vector faster.

Without Indexing ⇒ Search every item.

With Indexing ⇒ Only search relevant items.

Vector Indexing improves:

→ Speed

→ Scalability

→ query performance.

⇒ Similarity Based Searching : Nearest Neighbors.

Exact nearest neighbor search

↓
finding exact & accurate nearest vectors.

→ Very expensive at high scale.

Approximate nearest neighbor (ANN) search

→ Find nearest neighbors based on approximations.

→ Widely used.

Exact Search (vs) Similarity Search